ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. ECE ENGINEERING — SEMESTER 6

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Explain CMOS Inverter operation: DC transfer characteristics, Noise margins, and regions of operation. [14 Marks]

[OR]

Q.1 (Alt) Derive Drain current equations for nMOS and pMOS transistors in linear and saturation regions. [14 Marks]

--- UNIT II ---

Q.2 Explain CMOS Fabrication process: N-Well, P-Well, and Twin-Tub processes with photolithography steps. [14 Marks]

[OR]

Q.2 (Alt) Explain Lambda-based Design Rules and Stick Diagrams for CMOS NAND and NOR gates. [14 Marks]

--- UNIT III ---

Q.3 Explain Dynamic CMOS Logic: Precharge and Evaluate phases, Charge Sharing, and Domino Logic. [14 Marks]

[OR]

Q.3 (Alt) Compare Static CMOS, Pseudo-nMOS, and Pass Transistor Logic / Transmission Gates. [14 Marks]

--- UNIT IV ---

Q.4 Explain Semiconductor Memories: 6T SRAM cell read/write operation and 1T DRAM cell refresh cycle. [14 Marks]

[OR]

Q.4 (Alt) Describe Flash memory (NAND vs NOR) and ROM architectures. [14 Marks]

--- UNIT V ---

Q.5 Write Verilog HDL code for 4-bit Carry Lookahead Adder and Finite State Machines (FSM). [14 Marks]

[OR]

Q.5 (Alt) Explain ASIC Design Flow, FPGA architecture (CLBs, LUTs, IOBs), and Built-In Self-Test (BIST). [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****